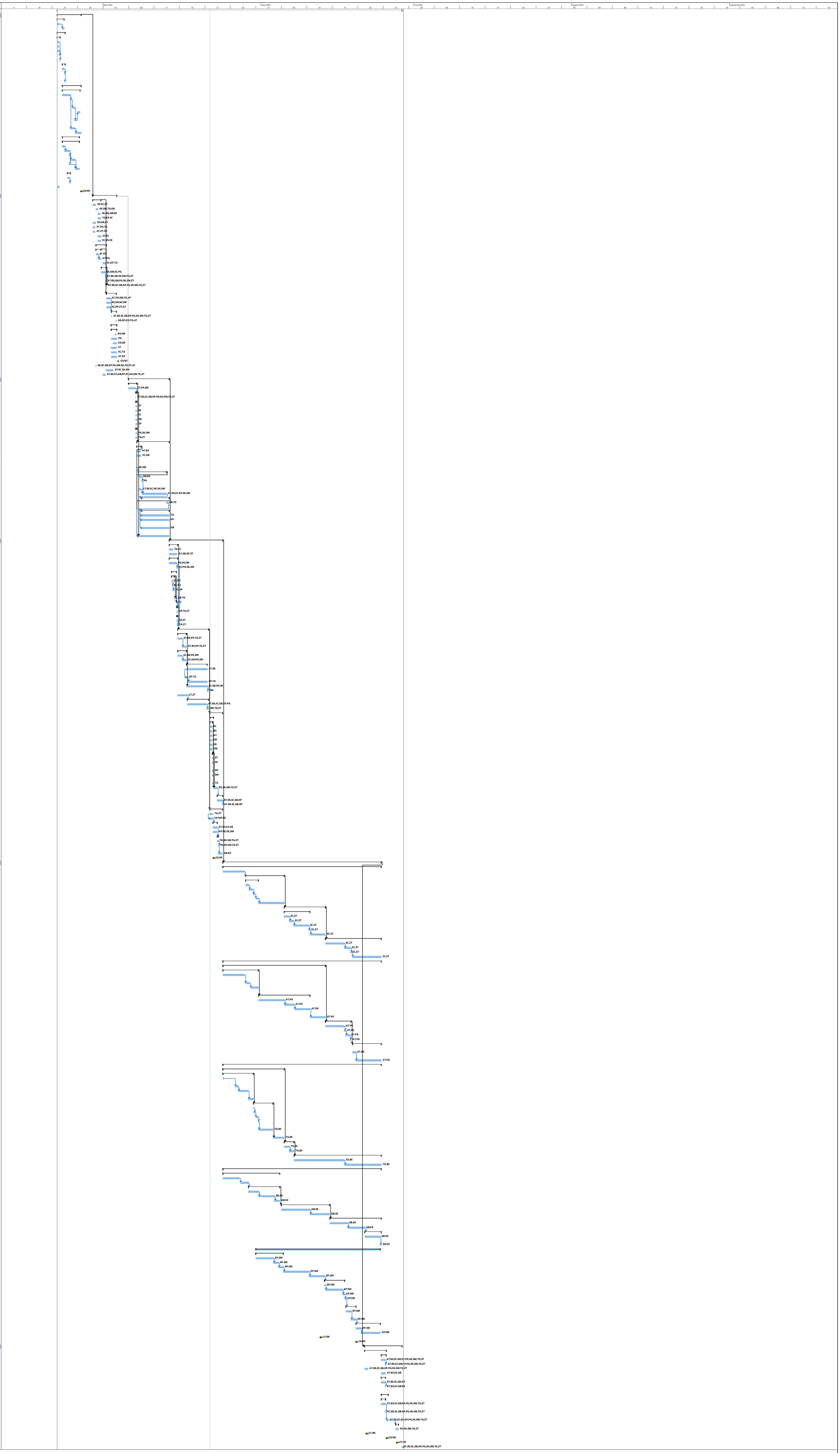
ID	WBS	Task Name	Resource Names	Duration	Work	Start	Predecessors	Notes
0		**Design Synthesis Exercise**		**376 hrs**	**3,670 hrs**	**Mon 22/04/2**		
1	**1**	**Project planning**		**40 hrs**	**370 hrs**	**Mon 22/04/24**		
2	**1.1**	**Define mission and project objectives**		**9 hrs**	**20 hrs**	**Mon 22/04/24**		
3	1.1.1	Mission Need Statement (MNS)		1 hr	10 hrs	Mon 22/04/24		
4	1.1.2	Project Objective Statement (POS)		1 hr	10 hrs	Tue 23/04/24	3	
5	**1.2**	**Group Organization**		**13 hrs**	**100 hrs**	**Mon 22/04/24**		
6	**1.2.1**	**Roles**		**5 hrs**	**70 hrs**	**Mon 22/04/24**		
7	1.2.1.1	Identify organizational roles		1 hr	10 hrs	Mon 22/04/24		
8	1.2.1.2	Belbin test		1 hr	10 hrs	Mon 22/04/24		
9	1.2.1.3	Identify technical roles		2 hrs	20 hrs	Mon 22/04/24		
10	1.2.1.4	Identify role responsibilities		1 hr	10 hrs	Mon 22/04/24	7,9	
11	1.2.1.5	Assign roles		2 hrs	20 hrs	Mon 22/04/24	10	
12	**1.2.2**	**Group organisation**		**5 hrs**	**30 hrs**	**Tue 23/04/24**		
13	1.2.2.1	Identify departments		1 hr	10 hrs	Tue 23/04/24		
14	1.2.2.2	Identify departments responsibilities		1 hr	10 hrs	Tue 23/04/24	13	
15	1.2.2.3	Organogram creation		3 hrs	10 hrs	Tue 23/04/24	14	
16	**1.3**	**Work division**		**32 hrs**	**192 hrs**	**Tue 23/04/24**		
17	**1.3.1**	**Work flow diagram**		**28 hrs**	**92 hrs**	**Tue 23/04/24**		
18	1.3.1.1	Determine tasks to be performed		1.5 days	40 hrs	Tue 23/04/24		
19	1.3.1.2	Determine relationships between tasks		0.5 days	20 hrs	Wed 24/04/24	18	
20	1.3.1.3	Determine optimal tasks order		2 hrs	4 hrs	Thu 25/04/24	19	
21	1.3.1.4	Assign tasks based on group organisation		0.5 days	8 hrs	Fri 26/04/24	22	
22	1.3.1.5	Estimate time required to perform task		6 hrs	20 hrs	Thu 25/04/24	20	
23	1.3.2	Work breakdown diagram		1 day	40 hrs	Wed 24/04/24	18	
24	1.3.3	Gantt chart		1.5 days	60 hrs	Thu 25/04/24	23	
25	**1.4**	**Risk management**		**25 hrs**	**33 hrs**	**Tue 23/04/24**		
26	**1.4.1**	**Create risk management protocol**		**25 hrs**	**33 hrs**	**Tue 23/04/24**		
27	1.4.1.1	Organisation SWOT analysis		3 hrs	6 hrs	Tue 23/04/24		
28	1.4.1.2	Identify risks		6 hrs	4 hrs	Tue 23/04/24	27	
29	1.4.1.3	Prioritise risks		3 hrs	6 hrs	Wed 24/04/24	28	
30	1.4.1.4	Risks mitigation		1 day	7 hrs	Wed 24/04/24	29	
31	1.4.1.5	Decide who is monitoring the risks		1 hr	2 hrs	Wed 24/04/24	28	
32	1.4.1.6	Create risk management schedule		5 hrs	8 hrs	Thu 25/04/24	31,30	
33	**1.5**	**Organisational approach to sustainability**		**5 hrs**	**5 hrs**	**Wed 24/04/24**		
34	1.5.1	Decide who is responsible		1 hr	1 hr	Wed 24/04/24		
35	1.5.2	Decide which methods will be used		4 hrs	4 hrs	Wed 24/04/24	34	
36	1.6	Team guidelines		0.25 days	20 hrs	Mon 22/04/24		
37	1.7	Draft Project Plan		0 days	0 hrs	Fri 26/04/24		
38	**2**	**Conceptual development phase**		**5 days**	**400 hrs**	**Mon 29/04/24**	**1**	
39	**2.1**	**Literature study**		**13.75 hrs**	**90 hrs**	**Mon 29/04/24**		
40	2.1.1	Existing eVTOL solutions	KP,EC,ZT	0.65 days	15 hrs	Mon 29/04/24		
41	2.1.2	Unfolding mechanism	SK,SM,TG,GB	0.31 days	10 hrs	Mon 29/04/24		
42	2.1.3	Noise reduction system	SK,SM,GB,BS	0.31 days	10 hrs	Tue 30/04/24		
43	2.1.4	Dynamic modelling and stability	TG,KP,EC	0.42 days	10 hrs	Tue 30/04/24		
44	2.1.5	Battery solutions	BS,GB,AT	0.63 days	15 hrs	Mon 29/04/24		
45	2.1.6	Material properties	ZT,PG,TG	0.3 days	5 hrs	Mon 29/04/24		
46	2.1.7	Production methods	AT,ZT,PG	0.21 days	5 hrs	Mon 29/04/24		
47	2.1.8	Recycling and reconversion	ZT,EC	0.63 days	10 hrs	Tue 30/04/24		
48	2.1.9	Ground operations	EC,BS,SK	0.42 days	10 hrs	Tue 30/04/24		
49	**2.2**	**Requirement determination**		**17 hrs?**	**75 hrs**	**Mon 29/04/24**		
50	**2.2.1**	**Stakeholder analysis**		**7.5 hrs?**	**15 hrs**	**Mon 29/04/24**		
51	2.2.1.1	Stakeholder identification	AT,PG	0.31 days?	5 hrs	Mon 29/04/24		
52	2.2.1.2	Stakeholder requirements	AT,PG	0.63 days?	10 hrs	Tue 30/04/24	51	
53	2.2.2	Determine mission profile	EC,KP,TG	0.42 days?	10 hrs	Wed 01/05/24		
54	**2.2.3**	**Technical requirements**		**6.79 hrs?**	**40 hrs**	**Tue 30/04/24**		
55	2.2.3.1	Technical mission requirements	SK,SM,EC,PG	0.31 days?	10 hrs	Tue 30/04/24		
56	2.2.3.2	System requirements	AT,BS,GB,SK,SM,PG,ZT	0.27 days?	15 hrs	Wed 01/05/24	55	
57	2.2.3.3	Sub-system requirements	AT,BS,GB,PG,SK,SM,ZT	0.27 days?	15 hrs	Wed 01/05/24	56	
58	2.2.4	Killer, driving and key requirements identification	AT,BS,EC,GB,KP,PG,SK,	0.13 days?	10 hrs	Wed 01/05/24	50,54	
59	**2.3**	**System concept generation**		**15 hrs?**	**60 hrs**	**Wed 01/05/24**	**39**	
60	2.3.1	Wing configuration	AT,PG,SM,TG,KP	0.38 days?	15 hrs	Wed 01/05/24		
61	2.3.2	Propulsion concept	BS,GB,SK,SM	0.31 days?	10 hrs	Wed 01/05/24		
62	2.3.3	Wing folding concept	EC,KP,ZT,AT	0.47 days?	15 hrs	Wed 01/05/24		
63	**2.3.4**	**Design option tree**		**7 hrs?**	**20 hrs**	**Thu 02/05/24**	**60,61,62**	
64	2.3.4.2	Eliminate unfeasible	AT,BS,EC,GB,KP,PG,SK,	0.13 days?	10 hrs	Thu 02/05/24		
65	2.3.4.1	Create diagram	GB,KP,SM,TG,AT	0.25 days?	10 hrs	Fri 03/05/24		
66	**2.4**	**Preliminary concepts evaluation**		**11 hrs?**	**85 hrs**	**Thu 02/05/24**		
67	**2.4.1**	**Performance analysis**		**10 hrs?**	**20 hrs**	**Thu 02/05/24**		
68	2.4.1.1	Evaluate folding mechanisms loads	BS,SM	0.63 days?	10 hrs	Fri 03/05/24		
69	2.4.1.2	Range estimates	PG	1.25 days?	10 hrs	Thu 02/05/24		
70	2.4.2	Sustainability analysis	SK,GB	0.94 days?	15 hrs	Fri 03/05/24		
71	2.4.3	Cost analysis	ZT	1.25 days?	10 hrs	Thu 02/05/24		
72	2.4.4	Risk analysis	EC,TG	1.25 days?	20 hrs	Thu 02/05/24		
73	2.4.5	Market research	AT,KP	1.25 days?	20 hrs	Thu 02/05/24		
74	2.5	Baseline report		0 days	0 hrs	Fri 03/05/24		
75	2.6	Formal meeting	BS,EC,GB,KP,PG,SM,SK,	0.25 days	20 hrs	Mon 29/04/24		
76	2.7	Quality assurance	AT,EC,SK,SM	1.25 days?	40 hrs	Wed 01/05/24		
77	2.8	Implementing feedback	AT,BS,EC,GB,KP,PG,SK,	0.38 days?	30 hrs	Wed 01/05/24		
78	**3**	**Trade-off phase**		**32 hrs**	**300 hrs**	**Mon 06/05/24**	**38**	
79	**3.1**	**Set-up trade-off**		**14 hrs**	**80 hrs**	**Mon 06/05/24**		
80	3.1.1	Assess existing trade-off strategies	AT,KP,SM	10 hrs	30 hrs	Mon 06/05/24		
81	**3.1.2**	**Come up with trade-off criteria**		**4 hrs**	**50 hrs**	**Tue 07/05/24**	**80**	
82	3.1.2.1	Brainstorm different criteria	AT,BS,EC,GB,KP,PG,SK,	1 hr	10 hrs	Tue 07/05/24		
83	**3.1.2.2**	**Assess technical aspects**		**4 hrs**	**20 hrs**	**Tue 07/05/24**		
84	3.1.2.2.1	Analyse technology readiness	AT	4 hrs	4 hrs	Tue 07/05/24		
85	3.1.2.2.2	Analyse transition efficiency	BS	4 hrs	4 hrs	Tue 07/05/24		
86	3.1.2.2.3	Evaluate structural complexity	EC	4 hrs	4 hrs	Tue 07/05/24		
87	3.1.2.2.4	Analyse noise produced	GB	4 hrs	4 hrs	Tue 07/05/24		
88	3.1.2.2.5	Evaluate energy consumption	KP	4 hrs	4 hrs	Tue 07/05/24		
89	**3.1.2.3**	**Assess non-technical aspects**		**4 hrs**	**20 hrs**	**Tue 07/05/24**		
90	3.1.2.3.1	Analyse safety	PG,SK,SM	4 hrs	12 hrs	Tue 07/05/24		
91	3.1.2.3.2	Analyse sustainability	TG,ZT	4 hrs	8 hrs	Tue 07/05/24		
92	**3.2**	**Performing trade-offs**		**18 hrs**	**220 hrs**	**Tue 07/05/24**	**79**	
93	**3.2.1**	**Give weights to each criteria**		**5 hrs**	**20 hrs**	**Tue 07/05/24**		
94	3.2.1.1	Decide upon weighing scale	AT,BS	3 hrs	6 hrs	Tue 07/05/24		
95	3.2.1.2	Research currently available technologies to use as references for the assignment of weights	EC,GB	5 hrs	10 hrs	Tue 07/05/24		
96	3.2.1.3	Justify assigned weights	SK,SM	2 hrs	4 hrs	Tue 07/05/24		
97	**3.2.2**	**Creating trade-off table**		**11 hrs**	**60 hrs**	**Wed 08/05/24**	**96**	
98	3.2.2.1	Research effective table layouts	GB,PG	5 hrs	10 hrs	Wed 08/05/24		
99	3.2.2.2	Converge on appropriate table layout	PG	2 hrs	2 hrs	Wed 08/05/24	98	
100	3.2.2.3	Assign weights to candidates	AT,BS,EC,KP,SK,SM	4 hrs	24 hrs	Wed 08/05/24		
101	3.2.2.4	Draw and compile table	AT,BS,EC,KP,SK,SM	4 hrs	24 hrs	Wed 08/05/24	99,100	
102	**3.2.3**	**Determine a design choice**		**10 hrs**	**40 hrs**	**Wed 08/05/24**	**101**	
103	3.2.3.1	Determine candidate with highest overall score	GB,TG	5 hrs	10 hrs	Mon 13/05/24		
104	**3.2.3.2**	**Validate result**		**10 hrs**	**30 hrs**	**Wed 08/05/24**	**103**	
105	3.2.3.2.1	Perform sensitivity analysis	TG	10 hrs	10 hrs	Wed 08/05/24	103	
106	3.2.3.2.2	Perform iterations on trade-off table if needed	KP	10 hrs	10 hrs	Wed 08/05/24	103	
107	3.2.3.2.3	Perform iterations on weights and criteria if needed	GB	10 hrs	10 hrs	Wed 08/05/24	103	
108	3.2.4	Report trade-off		16 hrs	100 hrs	Wed 08/05/24	103,81,93,97,102	
109	**4**	**Preliminary design**		**60 hrs**	**600 hrs**	**Tue 14/05/24**	**78**	
110	**4.1**	**Perform class I preliminary sizing**		**15 hrs**	**150 hrs**	**Tue 14/05/24**		
111	4.1.1	Estimate battery size	TG,EC	8 hrs	16 hrs	Tue 14/05/24		
112	4.1.2	Perform class I weight estimation	AT,GB,KP,ZT	12.5 hrs	50 hrs	Tue 14/05/24		
113	**4.1.3**	**Size engines and wing**		**15 hrs**	**50 hrs**	**Tue 14/05/24**		
114	4.1.3.1	Generate W/S-W/P diagrams	PG,SK,SM	10 hrs	30 hrs	Tue 14/05/24		
115	4.1.3.2	Choose design point	EC,PG,SK,SM	5 hrs	20 hrs	Wed 15/05/24	114	
116	**4.1.4**	**Generate payload range diagrams**		**7.5 hrs**	**16 hrs**	**Tue 14/05/24**		
117	**4.1.4.1**	**Calculate weights**		**5.5 hrs**	**11 hrs**	**Tue 14/05/24**		
118	4.1.4.1.1	Calculate payload	EC,PG	1.5 hrs	3 hrs	Tue 14/05/24		
119	4.1.4.1.2	Calculate fuel weight	EC,PG	2 hrs	4 hrs	Tue 14/05/24	118	
120	4.1.4.1.3	Calculate Maximum takeoff weight	EC,KP	2 hrs	4 hrs	Tue 14/05/24	119	
121	4.1.4.2	Identify weight range relations	GB,TG	2 hrs	4 hrs	Wed 15/05/24	117	
122	4.1.4.3	Draw diagram	TG	1 hr	1 hr	Wed 15/05/24	117,121FF	
123	**4.1.5**	**Identify load cases**		**4.5 hrs**	**18 hrs**	**Wed 15/05/24**	**122**	
124	4.1.5.1	Analyse gust loads	KP,TG,ZT	3.33 hrs	10 hrs	Wed 15/05/24		
125	**4.1.5.2**	**Generate flight envelope**		**4 hrs**	**8 hrs**	**Wed 15/05/24**		
126	4.1.5.2.1	Identify operational limits	KP,ZT	2 hrs	4 hrs	Wed 15/05/24		
127	4.1.5.2.2	Identify load factors	KP,ZT	2 hrs	4 hrs	Wed 15/05/24	126,115FF	
128	**4.2**	**Perform subsystem preliminary sizing**		**25 hrs**	**251 hrs**	**Wed 15/05/24**	**110**	
129	**4.2.1**	**Design unfolding mechanism**		**10 hrs**	**50 hrs**	**Wed 15/05/24**		
130	4.2.1.1	Identify unfolding mechanisms concepts	AT,BS,KP,TG,ZT	5 hrs	25 hrs	Wed 15/05/24		
131	4.2.1.2	Trade-off unfolding mechanisms	AT,BS,KP,TG,ZT	5 hrs	25 hrs	Thu 16/05/24	130	
132	**4.2.2**	**Design power system**		**10 hrs**	**40 hrs**	**Wed 15/05/24**		
133	4.2.2.1	Size accumulator	EC,GB,PG,SM	5 hrs	20 hrs	Wed 15/05/24		
134	4.2.2.2	Size power distribution system	EC,GB,PG,SM	5 hrs	20 hrs	Thu 16/05/24	133	
135	**4.2.3**	**Size wing**		**10 hrs**	**40 hrs**	**Fri 17/05/24**	**129**	
136	4.2.3.1	Estimate realistic aerodynamic coefficients	AT,BS	10 hrs	20 hrs	Fri 17/05/24		
137	4.2.3.2	Approximate aspect ratio	KP,TG	5 hrs	10 hrs	Fri 17/05/24	136SS	
138	4.2.3.3	Determine wingspan	KP,TG	5 hrs	10 hrs	Fri 17/05/24	137	
139	4.2.4	Size propulsion subsystem	EC,GB,PG,SK	10 hrs	40 hrs	Fri 17/05/24	132	
140	4.2.5	Select autonomous control	SM	4 hrs	4 hrs	Tue 21/05/24	139	
141	4.2.6	Design and size other subsystems	AT,ZT	13 hrs	26 hrs	Wed 15/05/24		
142	**4.2.7**	**Perform subsystem trade-offs**		**12 hrs**	**51 hrs**	**Fri 17/05/24**	**141**	
143	4.2.7.1	Create diagram	AT,BS,EC,GB,KP,PG	5 hrs	30 hrs	Fri 17/05/24		
144	4.2.7.2	Eliminate unfeasible concepts	SM,TG,ZT	7 hrs	21 hrs	Tue 21/05/24	143	
145	**4.3**	**Perform class II sizing**		**20 hrs**	**99 hrs**	**Wed 22/05/24**	**128**	
146	**4.3.1**	**Perform class II weight estimation**		**7 hrs**	**34 hrs**	**Wed 22/05/24**		
147	**4.3.1.1**	**Estimate masses of subsystems**		**3 hrs**	**14 hrs**	**Wed 22/05/24**		
148	4.3.1.1.1	Estimate fuselage mass	EC	2 hrs	2 hrs	Wed 22/05/24		
149	4.3.1.1.2	Estimate empenage mass	BS	2 hrs	2 hrs	Wed 22/05/24		
150	4.3.1.1.3	Estimate propeller mass	AT	2 hrs	2 hrs	Wed 22/05/24		
151	4.3.1.1.4	Estimate landing gear mass	GB	2 hrs	2 hrs	Wed 22/05/24		
152	4.3.1.1.5	Estimate wings mass	SK	3 hrs	3 hrs	Wed 22/05/24		
153	4.3.1.1.6	Estimate battery mass	SM	3 hrs	3 hrs	Wed 22/05/24		
154	**4.3.1.2**	**Estimate hinge mass**		**4 hrs**	**20 hrs**	**Wed 22/05/24**	**147**	
155	4.3.1.2.1	Research hinges on aircrafts	ZT	4 hrs	4 hrs	Wed 22/05/24		
156	4.3.1.2.2	Evaluate different designs in the market	SK	4 hrs	4 hrs	Wed 22/05/24		
157	4.3.1.2.3	Compare weight with literature	PG	4 hrs	4 hrs	Wed 22/05/24		
158	4.3.1.2.4	Ensure performance is within requirements	SM	4 hrs	4 hrs	Wed 22/05/24		
159	4.3.1.2.5	Estimate wing mass	TG	4 hrs	4 hrs	Wed 22/05/24		
160	4.3.2	Compare class I weight estimation with class II weight estimation	PG,SK,SM,TG,ZT	3 hrs	15 hrs	Wed 22/05/24	154,147	
161	**4.3.3**	**Iterate Class I preliminary sizing**		**10 hrs**	**50 hrs**	**Thu 23/05/24**	**160**	
162	4.3.3.1	Create iteration tool	AT,BS,EC,GB,KP	8 hrs	40 hrs	Thu 23/05/24		
163	4.3.3.2	Analyse iteration	AT,BS,EC,GB,KP	2 hrs	10 hrs	Fri 24/05/24	162	
164	**4.4**	**Assess stability and control**		**19 hrs**	**100 hrs**	**Wed 22/05/24**	**128**	
165	4.4.1	Generate loading diagram	TG,ZT	5 hrs	10 hrs	Wed 22/05/24		
166	4.4.2	Determine center of gravity range	SK,SM,TG	5 hrs	15 hrs	Wed 22/05/24	165SS	
167	**4.4.3**	**Generate scissor plot**		**5 hrs**	**40 hrs**	**Wed 22/05/24**	**166**	
168	4.4.3.1	Determine stability curve	AT,BS,EC,GB	5 hrs	20 hrs	Wed 22/05/24		
169	4.4.3.2	Determine controllability curve	KP,PG,SK,SM	5 hrs	20 hrs	Wed 22/05/24		
170	**4.4.4**	**Size tail**		**5 hrs**	**25 hrs**	**Thu 23/05/24**	**168**	
171	4.4.4.1	Determine aerodynamic forces	PG,SK,SM,TG,ZT	4 hrs	20 hrs	Thu 23/05/24		
172	4.4.4.2	Check compliance with engine inoperative case	PG,SK,SM,TG,ZT	1 hr	5 hrs	Thu 23/05/24	171	
173	4.4.5	Iterate class I and class II procedures	GB,KP	5 hrs	10 hrs	Thu 23/05/24	171	
174	4.5	Midterm report		0 days	0 hrs	Wed 22/05/24		
175	**5**	**Detailed design**		**170 hrs**	**1,700 hrs**	**Fri 24/05/24**	**109**	
176	**5.1**	**Perform stability analysis**		**170 hrs**	**360 hrs**	**Fri 24/05/24**		
177	5.1.1	Analyse ground stability		20 hrs	50 hrs	Fri 24/05/24		KP[50%],EC,ZT
178	**5.1.2**	**Analyse transition stability**		**45 hrs**	**100 hrs**	**Wed 29/05/24**	**177**	
179	**5.1.2.1**	**Model dynamic transition**		**20 hrs**	**50 hrs**	**Wed 29/05/24**		
180	5.1.2.1.1	Derive dynamic model		6 hrs	15 hrs	Wed 29/05/24		KP[50%],EC,ZT
181	5.1.2.1.2	Implement dynamic model		4 hrs	10 hrs	Wed 29/05/24	180	KP[50%],EC,ZT
182	5.1.2.1.3	V&V the program		6 hrs	15 hrs	Thu 30/05/24	181	KP[50%],EC,ZT
183	5.1.2.1.4	Perform simulations		4 hrs	10 hrs	Fri 31/05/24	182	KP[50%],EC,ZT
184	5.1.2.2	Analyse stability of dynamic model		25 hrs	50 hrs	Fri 31/05/24	183	KP[50%],EC,ZT
185	**5.1.3**	**Analyse in flight stability**		**50 hrs**	**100 hrs**	**Wed 05/06/24**	**178**	
186	**5.1.3.1**	**Model dynamics in cruise condition**		**25 hrs**	**50 hrs**	**Wed 05/06/24**		
187	5.1.3.1.1	Derive dynamic model	EC,ZT	7.5 hrs	15 hrs	Wed 05/06/24		
188	5.1.3.1.2	Implement dynamic model	EC,ZT	5 hrs	10 hrs	Thu 06/06/24	187	
189	5.1.3.1.3	V&V the program	EC,ZT	7.5 hrs	15 hrs	Fri 07/06/24	188	
190	5.1.3.1.4	Perfrom simulations	EC,ZT	5 hrs	10 hrs	Mon 10/06/24	189	
191	5.1.3.2	Analyse stabilty of dynamic model	EC,ZT	25 hrs	50 hrs	Mon 10/06/24	190	
192	**5.1.4**	**Design controller**		**55 hrs**	**110 hrs**	**Thu 13/06/24**	**185**	
193	5.1.4.1	Identify possible controller types	EC,ZT	10 hrs	20 hrs	Thu 13/06/24		
194	5.1.4.2	List advantages and disadvantages	EC,ZT	12.5 hrs	25 hrs	Mon 17/06/24	193	
195	5.1.4.3	Select controller type	EC,ZT	2.5 hrs	5 hrs	Tue 18/06/24	194	
196	5.1.4.4	Implement and tune controller	EC,ZT	30 hrs	60 hrs	Wed 19/06/24	195	
197	**5.2**	**Perform structural analysis**		**170 hrs**	**360 hrs**	**Fri 24/05/24**		
198	**5.2.1**	**Design hinge mechanism**		**115 hrs**	**250 hrs**	**Fri 24/05/24**		
199	**5.2.1.1**	**Identify hinge load cases**		**40 hrs**	**100 hrs**	**Fri 24/05/24**		
200	5.2.1.1.1	Develop tools for load calculation		20 hrs	50 hrs	Fri 24/05/24		KP[50%],AT,PG
201	5.2.1.1.2	Identify load cases		8 hrs	20 hrs	Wed 29/05/24	200	KP[50%],AT,PG
202	5.2.1.1.3	Compute internal loads for load cases		12 hrs	30 hrs	Thu 30/05/24	201	KP[50%],AT,PG
203	**5.2.1.2**	**Design hinge**		**50 hrs**	**100 hrs**	**Fri 31/05/24**	**199**	
204	5.2.1.2.1	Perform mechanical sizing	AT,PG	25 hrs	50 hrs	Fri 31/05/24		
205	5.2.1.2.2	Perform structural sizing	AT,PG	15 hrs	30 hrs	Wed 05/06/24	204	
206	5.2.1.2.3	Perform dynamic simulation on the design and iterate	AT,PG	10 hrs	20 hrs	Fri 07/06/24	205	
207	5.2.1.3	Select actuators	AT,PG	25 hrs	50 hrs	Mon 10/06/24	206	
208	**5.2.2**	**Material selection**		**25 hrs**	**50 hrs**	**Thu 13/06/24**	**198**	
209	5.2.2.1	Research competition materials	AT,PG	10 hrs	20 hrs	Thu 13/06/24		
210	5.2.2.2	Document material properties	AT,PG	5 hrs	10 hrs	Mon 17/06/24	209	
211	5.2.2.3	Analyse material performance	AT,PG	5 hrs	10 hrs	Mon 17/06/24	210	
212	5.2.2.4	Select material	AT,PG	5 hrs	10 hrs	Tue 18/06/24	211	
213	**5.2.3**	**Perform structural analysis of sub-systems**		**30 hrs**	**60 hrs**	**Wed 19/06/24**	**208**	
214	5.2.3.1	Check compliance with predicted loads	AT,PG	5 hrs	10 hrs	Wed 19/06/24		
215	5.2.3.2	Perform iteration	AT,PG	25 hrs	50 hrs	Wed 19/06/24	214	
216	**5.3**	**Aerodynamic analysis**		**170 hrs**	**360 hrs**	**Fri 24/05/24**		
217	**5.3.1**	**Finalise wing design**		**65 hrs**	**150 hrs**	**Fri 24/05/24**		
218	**5.3.1.1**	**Select airfoil**		**32 hrs**	**80 hrs**	**Fri 24/05/24**		
219	5.3.1.1.1	Build airfoil model in simulation enviroment		4 hrs	10 hrs	Fri 24/05/24		SM[50%],TG,BS
220	5.3.1.1.2	Perform simulation		4 hrs	10 hrs	Mon 27/05/24	219	SM[50%],TG,BS
221	5.3.1.1.3	Analyse aerodynamic characteristics		16 hrs	40 hrs	Mon 27/05/24	220	SM[50%],TG,BS
222	5.3.1.1.4	Choose airfoil		8 hrs	20 hrs	Wed 29/05/24	221	SM[50%],TG,BS
223	**5.3.1.2**	**Design wing control surface**		**13 hrs**	**30 hrs**	**Thu 30/05/24**	**218**	
224	5.3.1.2.1	Estimate roll rate		2 hrs	5 hrs	Thu 30/05/24		SM[50%],TG,BS
225	5.3.1.2.2	Establish initial dimentions		2 hrs	5 hrs	Thu 30/05/24	224	SM[50%],TG,BS
226	5.3.1.2.3	Determine likely deflection		2 hrs	5 hrs	Fri 31/05/24	225	SM[50%],TG,BS
227	5.3.1.2.4	Analyse aerodynamic parameters		2 hrs	5 hrs	Fri 31/05/24	226	SM[50%],TG,BS
228	5.3.1.2.5	Iterate dimensions and establish final design	TG,BS	5 hrs	10 hrs	Fri 31/05/24	227	
229	5.3.1.3	Design main wing planform	TG,BS	20 hrs	40 hrs	Mon 03/06/24	223	
230	**5.3.2**	**Design fuselage shape**		**15 hrs**	**30 hrs**	**Wed 05/06/24**	**217**	
231	5.3.2.1	Design nose cone	TG,BS	7.5 hrs	15 hrs	Wed 05/06/24		
232	5.3.2.2	Design tail cone	TG,BS	7.5 hrs	15 hrs	Thu 06/06/24	231	
233	**5.3.3**	**Perform aerodynamic simulations**		**90 hrs**	**180 hrs**	**Fri 07/06/24**	**230**	
234	5.3.3.1	Calculate lift and drag	TG,BS	45 hrs	90 hrs	Fri 07/06/24		
235	5.3.3.2	Determine stability coefficents	TG,BS	45 hrs	90 hrs	Mon 17/06/24	234	
236	**5.4**	**Perform propulsion analysis**		**170 hrs**	**360 hrs**	**Fri 24/05/24**		
237	**5.4.1**	**Design engines**		**60 hrs**	**140 hrs**	**Fri 24/05/24**		
238	5.4.1.1	Study off-the-shelf engines		12 hrs	30 hrs	Fri 24/05/24		SM[50%],GB,SK
239	5.4.1.2	Design propellers		12 hrs	30 hrs	Tue 28/05/24	238	SM[50%],GB,SK
240	**5.4.1.3**	**Analyse performance**		**36 hrs**	**80 hrs**	**Wed 29/05/24**	**239**	
241	5.4.1.3.1	Perform thrust calculations		16 hrs	40 hrs	Wed 29/05/24		SM[50%],GB,SK
242	5.4.1.3.2	Determine drag characteristics	GB,SK	10 hrs	20 hrs	Fri 31/05/24	241	
243	5.4.1.3.3	Evaluate wake of engines	GB,SK	10 hrs	20 hrs	Mon 03/06/24	242	
244	**5.4.2**	**Analyse noise pollution of aircraft**		**60 hrs**	**120 hrs**	**Wed 05/06/24**	**240**	
245	5.4.2.1	Analyse ground noise	GB,SK	30 hrs	60 hrs	Wed 05/06/24		
246	5.4.2.2	Analyse cabin noise	GB,SK	30 hrs	60 hrs	Mon 10/06/24	245	
247	**5.4.3**	**Design power system**		**50 hrs**	**100 hrs**	**Fri 14/06/24**	**244**	
248	5.4.3.1	Perform energy accumulator design	GB,SK	12.5 hrs	25 hrs	Fri 14/06/24		
249	5.4.3.2	Design power distribution	GB,SK	25 hrs	50 hrs	Tue 18/06/24	248	
250	**5.4.3.3**	**Design voltage conversion system**		**12.5 hrs**	**25 hrs**	**Fri 21/06/24**	**249**	
251	5.4.3.3.1	Determine voltage needed by each subsystem	GB,SK	7.5 hrs	15 hrs	Fri 21/06/24		
252	5.4.3.3.2	Determine wiring needed	GB,SK	5 hrs	10 hrs	Mon 24/06/24	251	
253	**5.5**	**Implement 3D model**		**130 hrs**	**260 hrs**	**Fri 31/05/24**		
254	**5.5.1**	**Model parts**		**25 hrs**	**50 hrs**	**Fri 31/05/24**		
255	5.5.1.1	Model mechanism part	KP,SM	10 hrs	20 hrs	Fri 31/05/24		
256	5.5.1.2	Model propeller	KP,SM	7.5 hrs	15 hrs	Mon 03/06/24	255	
257	5.5.1.3	Model other key parts	KP,SM	7.5 hrs	15 hrs	Tue 04/06/24	256	
258	5.5.2	Model sub-assembly	KP,SM	25 hrs	50 hrs	Wed 05/06/24	257	
259	5.5.3	Complete assembly design	KP,SM	25 hrs	50 hrs	Mon 10/06/24	258	
260	**5.5.4**	**3D printing the model**		**15 hrs**	**30 hrs**	**Thu 13/06/24**	**259**	
261	5.5.4.1	Slice the model	KP,SM	2.5 hrs	5 hrs	Thu 13/06/24		
262	5.5.4.2	Print the 3D model	KP,SM	10 hrs	20 hrs	Thu 13/06/24	261	
263	5.5.4.3	Check 3D print quality	KP,SM	2.5 hrs	5 hrs	Mon 17/06/24	262	
264	5.5.5	Perform partial class III weight estimation	KP,SM	5 hrs	10 hrs	Mon 17/06/24	263	
265	**5.5.6**	**Perform Verification**		**15 hrs**	**30 hrs**	**Mon 17/06/24**	**264**	
266	5.5.6.1	Perform mathematical demonstration	KP,SM	7.5 hrs	15 hrs	Mon 17/06/24		
267	5.5.6.2	Test software enviroment	KP,SM	7.5 hrs	15 hrs	Tue 18/06/24	266	
268	**5.5.7**	**Perform Validation**		**20 hrs**	**40 hrs**	**Wed 19/06/24**	**267**	
269	5.5.7.1	Compare with other design	KP,SM	7.5 hrs	15 hrs	Wed 19/06/24		
270	5.5.7.2	Validation with existing models	KP,SM	12.5 hrs	25 hrs	Thu 20/06/24	269	
271	5.6	Jury Summary		0 days	0 hrs	Wed 12/06/24		
272	5.7	Draft final report		0 days	0 hrs	Wed 19/06/24		
273	**6**	**Project finalisation**		**47 hrs**	**300 hrs**	**Fri 21/06/24**	**175**	
274	**6.1**	**Attend symposium**		**20.5 hrs**	**165 hrs**	**Fri 21/06/24**		
275	**6.1.1**	**Present final presentation**		**7.5 hrs**	**75 hrs**	**Mon 24/06/24**		
276	6.1.1.1	Plan presentation	AT,BS,EC,GB,KP,PG,SK,	3 hrs	30 hrs	Mon 24/06/24		
277	6.1.1.2	Rehearse presentation	AT,BS,EC,GB,KP,PG,SK,	4.5 hrs	45 hrs	Tue 25/06/24	276	
278	6.1.2	Create poster	AT,BS,EC,GB,KP,PG,SK,	7 hrs	70 hrs	Fri 21/06/24		
279	6.1.3	Submit jury summary	AT,BS,EC,GB	5 hrs	20 hrs	Mon 24/06/24		
280	**6.2**	**Attend final review**		**5 hrs**	**25 hrs**	**Mon 24/06/24**		
281	6.2.1	Submit agenda	AT,BS,EC,GB,KP	4 hrs	20 hrs	Mon 24/06/24		
282	6.2.2	Send out appropriate guest invitations	AT,BS,EC,GB,KP	1 hr	5 hrs	Tue 25/06/24	281	
283	**6.3**	**Final report**		**10 hrs**	**75 hrs**	**Mon 24/06/24**		
284	**6.3.1**	**Close-out project**		**5.5 hrs**	**55 hrs**	**Mon 24/06/24**		
285	6.3.1.1	Obtain approval of the detailed design from PM	AT,BS,EC,GB,KP,PG,SK,	5 hrs	50 hrs	Mon 24/06/24		
286	6.3.1.2	Obtain approval of detailed design from customer	AT,BS,EC,GB,KP,PG,SK,	0.5 hrs	5 hrs	Tue 25/06/24	285	
287	6.3.2	Update executive overview	AT,BS,EC,GB,KP,PG,SK,	2 hrs	20 hrs	Tue 25/06/24	286	
288	**6.4**	**Report in "Red booklet"**		**5 hrs**	**25 hrs**	**Thu 27/06/24**	**287**	
289	6.4.1	Summerise project	PG,SK,SM,TG,ZT	5 hrs	25 hrs	Thu 27/06/24		
290	6.5	Poster		0 hrs	0 hrs	Fri 21/06/24		
291	6.6	Final Report Revision		0 hrs	0 hrs	Tue 25/06/24		
292	6.7	Red booklet		0 hrs	0 hrs	Thu 27/06/24		
293	6.8	Symposium presentation	AT,BS,EC,GB,KP,PG,SK,	1 hr	10 hrs	Fri 28/06/24		



October 2024
28 | 23 | 28 | 03 | 08 | 13 | 18 | 23 | 28

November 2024
02 | 07 | 12 | 17 | 22 | 27

December 2024
02 | 07 | 12 | 17 | 22 | 27

January 2025
01 | 06 | 11 | 16 | 21 | 26

February 2025
01 | 05 | 10 | 15 | 20 | 25

March 2025
02 | 07 | 12 | 17 | 22 | 27

April 2025
01

Design Synthesis Exercise
Wed 22/05/24

Task | Split | Milestone | Summary | Project Summary | Inactive Task | Inactive Milestone | Inactive Summary | Manual Task | Duration-only | Manual Summary Rollup | Manual Summary | Start-only | Finish-only | External Tasks | External Milestone | Deadline | Progress | Manual Progress